UNIT 2: DATA LITERACY

2.1.1 Introduction: Why Do We Need Data?

Imagine you are planning a school picnic. To decide the destination, you ask your classmates for their choices, check the weather forecast, calculate the total cost, and count how many students will participate. All these facts and figures help you take a better decision. Such useful information is called **data**.

In today's digital world, data is created almost all the time. Every photo we click, message we send, video we watch, online search we make, or marks we score in exams generates data. This data is stored, processed, and analyzed by computers.

Artificial Intelligence (AI) systems depend completely on data. Without data, AI cannot learn, think, or make decisions. Therefore, understanding data and knowing how to use it properly is an important skill for every student.

In this unit, you will learn what data is, why data literacy is important, how data is collected, processed, interpreted, and how different types of data are used in Artificial Intelligence.

2.1 Basics of Data Literacy

What is Data Literacy?

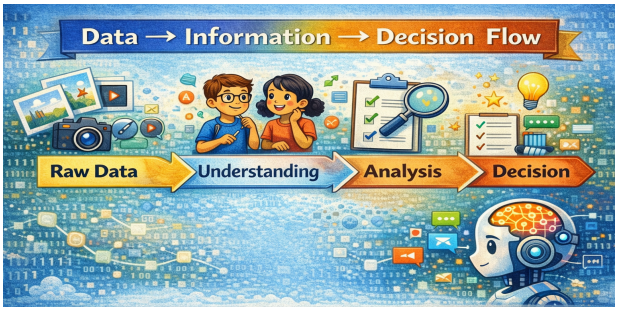
Data literacy is the ability to read, understand, analyze, and use data in a meaningful and responsible way.

A data-literate person is able to:

- Understand what data represents
- Identify useful and irrelevant data
- Analyze data correctly
- Interpret results meaningfully
- Use data in an ethical and responsible manner

In simple words,

“ Data literacy means being smart and careful while working with data” .



Why is Data Literacy Important?

Data literacy is important for many reasons:

- Artificial Intelligence systems learn only from data
- Incorrect or incomplete data can lead to wrong decisions
- Data helps us identify patterns, trends, and relationships
- Decisions based on data are usually more accurate than guesses

Example:
If a school uses incorrect attendance data, some students may be wrongly marked as defaulters. Accurate data helps schools take fair and correct decisions

Becoming Data Literate

To become data literate, a person should:

- Ask questions about the data
- Check where the data comes from
- Understand what the data is actually showing
- Avoid making quick assumptions
- Use data ethically and responsibly

Think About It:
When you read a headline like *“90% students prefer online learning”*, ask yourself:

- How many students were surveyed?
- Who conducted the survey?
- Is the data reliable and unbiased?
- Is the data reliable?

These questions help you think critically about data

Data Security and Privacy

Data security refers to protecting data from theft, misuse, or unauthorized access.

Data privacy means protecting personal information so that it is not shared without permission.

Personal data may include:

- Name
- Phone number
- Address
- Photographs
- Location information

AI systems often use large amounts of personal data. Therefore, it is very important to handle such data carefully and responsibly.

Best Practices for Cyber Safety

To stay safe in the digital world, students should follow these practices:

- Do not share personal details on unknown websites
- Use strong and unique passwords
- Avoid clicking on unknown or suspicious links
- Always log out from shared or public devices
- Read app permissions before allowing access

Remember:

> " Once data is shared online, it is very difficult to take it back" .

2.2 Acquiring, Processing, and Interpreting Data

What is Data Acquisition?

Data acquisition is the process of collecting data for a specific purpose.

In Artificial Intelligence projects, data is collected so that machines can learn patterns from it and make predictions or decisions. The quality of data collected directly affects the performance of an AI system.

Example:
To predict students' academic performance, data such as attendance, study hours, and previous exam marks may be collected.

Types of Data

Based on the form in which information is available, data can be classified into different types. Understanding these types is important because **different AI techniques are used for different types of data**.

The main types of data are:

- Numerical Data
- Categorical Data
- Text Data
- Image and Video Data

1. Numerical Data

Numerical data refers to information that is expressed using **numbers**. This type of data can be counted or measured and is often used for calculations, comparisons, and analysis.

Numerical data helps us find answers to questions such as:

- How many items are there?
- How much value is involved?
- How frequently does something happen?

Examples of Numerical Data include:

- Marks scored by students in examinations
- Height and weight measurements
- Temperature readings of a place
- Number of hours spent studying
- Speed of vehicles

Numerical data is further classified into two types. **Discrete data** consists of countable values such as the number of students in a class or the number of books in a library. **Continuous data**, on the other hand, includes measurable values like height, weight, and temperature.

Use in Artificial Intelligence:
AI systems rely on numerical data to perform calculations, detect trends, and make predictions. For example, numerical data is used in weather forecasting and in predicting students' academic performance.

2. Categorical Data

Categorical data is used to describe information that falls into **groups or categories**. Instead of numbers, this type of data represents qualities or labels.

Categorical data helps answer questions such as:

- Which category does it belong to?
- What type is it?
- Which group does it fall under?

Examples of Categorical Data include:

- Blood groups such as A, B, AB, and O
- Colours like red, blue, or green
- Gender categories such as male, female, or other
- Modes of transport like bus, train, or bicycle

Use in Artificial Intelligence:
AI systems use categorical data for classification tasks. This includes sorting emails, identifying different customer types, or grouping students based on performance levels.

3. Text Data

Text data consists of information written in the form of **words and sentences**. This type of data often carries meaning, opinions, and emotions, making it more complex to process.

Examples of Text Data include:

- Emails and text messages
- Online reviews and feedback
- News articles
- Comments on social media platforms

Use in Artificial Intelligence:
Text data is analyzed using **Natural Language Processing (NLP)**. NLP allows AI systems to understand human language and is used in chatbots, language translation apps, voice assistants, and sentiment analysis tools.

4. Image and Video Data

Image and video data contains information in **visual form**. Images are still pictures, while videos are made up of a sequence of images displayed over time.

Examples of Image and Video Data include:

- Photographs taken using cameras or phones
- CCTV surveillance recordings
- Medical imaging such as X-rays or scans

- Traffic monitoring videos

Use in Artificial Intelligence:
This type of data is processed using **Computer Vision**. Computer Vision helps AI systems recognize faces, identify objects, understand activities, and detect visual patterns.

Key Takeaway

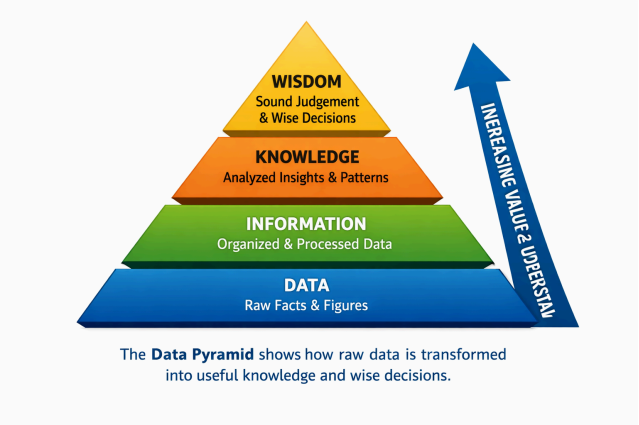
> Artificial Intelligence performs effectively only when the appropriate type of data is collected, understood, and used in the correct way.

Data Pyramid

The **Data Pyramid** is a framework that explains how raw facts are converted step by step into meaningful understanding and wise decisions. It highlights that data alone has very little value, but when it is properly processed, analyzed, and interpreted, it becomes useful for decision-making.

The Data Pyramid is made up of **four levels**, arranged from bottom to top:

- Data
- Information
- Knowledge
- Wisdom



As we move upward in the pyramid, the **value and understanding of data increase**.

1. Data (Bottom Level)

At the base of the pyramid lies **data**, which consists of raw and unprocessed facts. At this stage, data does not convey any clear meaning on its own.

Data can appear in different forms such as:

- Numbers
- Words
- Images
- Symbols

Example:
75, 82, 68, 90

These numbers by themselves do not explain anything unless we are told what they represent.

Daily Life Example:
Marks written on a sheet of paper without mentioning the subject, student name, or exam details are simply data.

2. Information

When raw data is **organized, arranged, or given proper context**, it turns into **information**. Information helps answer basic questions like *what*, *where*, and *when*.

Example:
"The marks scored by a student in Mathematics are 75, 82, 68, and 90."

Now the numbers are meaningful because we know what they refer to.

Role in Artificial Intelligence:
AI systems process raw data to convert it into information that can be understood and further analyzed.

3. Knowledge

Knowledge is formed when information is carefully examined and interpreted. At this stage, patterns, trends, and relationships become visible.

Knowledge helps answer questions such as:

- How does something happen?
- Why does it happen?

Example:
"Students who study regularly tend to score higher marks in Mathematics."

This conclusion is drawn by analyzing information and identifying a pattern.

Role in Artificial Intelligence:
AI systems use knowledge gained from past data to learn, adapt, and improve their performance over time.

4. Wisdom (Top Level)

At the top of the pyramid is **wisdom**, which involves using knowledge to take **correct, responsible, and thoughtful decisions**. Wisdom includes experience, judgment, and ethical understanding.

Wisdom answers the question:

- What should be done next?

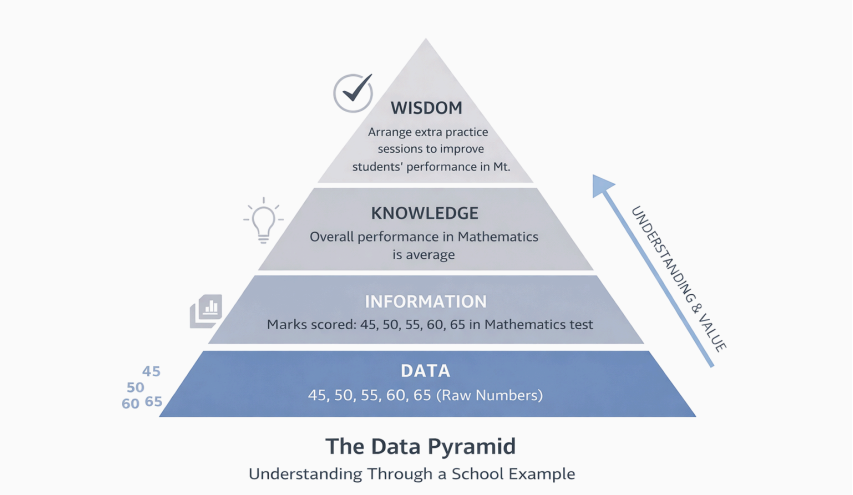
Example:
"Students should practice Mathematics daily to improve their performance."

This is a decision made by applying knowledge gained from data and information.

Role in Artificial Intelligence:
AI systems use wisdom-level insights to make recommendations, predict outcomes, and support decision-making.

School-Based Example of the Data Pyramid

The Data Pyramid can be easily understood using a simple school example. At the lowest level, we have raw data such as the numbers 45, 50, 55, 60, and 65. These numbers alone have no meaning because we do not know what they represent. When the same numbers are identified as the marks scored by students in a Mathematics test, they become information. By analyzing this information, we observe that most students have scored between 50 and 65 marks, indicating that the overall performance in Mathematics is average. This understanding represents knowledge. Finally, when this knowledge is used to make a decision, such as arranging extra practice sessions to improve students' performance, it reaches the level of wisdom. Thus, the Data Pyramid explains how raw data is transformed into information, knowledge, and finally wisdom for effective decision-making.



2.1.2 Why is the Data Pyramid Important?

The Data Pyramid helps us understand how raw facts are converted into meaningful decisions. It explains that data on its own has very little value unless it is properly processed and analyzed. As data moves upward through different levels, it becomes more meaningful and useful.

The Data Pyramid shows that:

- Raw data by itself does not provide clear meaning
- Processing and analysis help data gain context and understanding
- Good decisions are made using knowledge, not raw data
- Artificial Intelligence systems also follow this process to learn and act

Data becomes powerful only when it is transformed into information, knowledge, and wisdom.

Activity: Impact of Data Literacy in Online Shopping

(Choose any trending product)

Session Preparation and Logistics

- **Class Strength:** 40 students
- **Activity Type:** Pair Activity
- **Learning Mode:** Activity-based learning

Materials Required

Item	Quantity
Online shopping websites / product links	Not Applicable
Computers	20

Purpose of the Activity

The purpose of this activity is to make students aware of the importance of data literacy in online shopping. In today's digital world, many products are advertised with heavy discounts and attractive offers, which may not always be genuine. Students often depend on online information while purchasing products, so it is important for them to understand how to read and analyze data carefully.

Through this activity, students learn to examine prices, discount percentages, product descriptions, ratings, and customer reviews. By comparing information from different online platforms, they develop the ability to verify data, identify misleading information, and avoid online fraud. This process encourages logical thinking and responsible use of digital information in daily life.

Activity Brief (Pair Activity)

Students will work in pairs and explore popular online shopping websites to identify a trending product that is shown with a high discount. They will search for the same product on multiple platforms and compare the details provided by each source.

During this process, students will observe how the product is described, note differences in pricing, study customer ratings and reviews, and analyze important numerical figures. Based on their observations, they will decide whether the offer appears genuine or misleading and record their findings systematically.

Observation Table

Author of the Source	Weblink to the Source	How the Product or Offer Was Described	Key Figures (Price, Discount, Rating, etc.)

Ranking of Data Sources

After completing the comparison, students will rank the data sources from most reliable to least reliable. The ranking should be based on the accuracy of information, clarity of product details, authenticity of reviews, and consistency of pricing. Students must provide suitable remarks to justify their ranking.

Rank	Data Source	Remarks

By completing this activity, students understand that attractive discounts and promotional offers should not be accepted at face value. Online data can be presented in different ways, and careful analysis is necessary to understand its true meaning. The activity highlights how data literacy helps individuals compare multiple sources, evaluate reliability, and make informed decisions. It strengthens critical thinking skills and encourages safe and responsible online shopping habits.

Overall Impact of Data Literacy

Data literacy plays a vital role in today's digital and information-driven society. It helps individuals understand data, use it responsibly, and take informed decisions. Its impact can be seen in education, technology, and Artificial Intelligence.

1. Better Decision-Making

Data literacy allows people to base decisions on facts rather than assumptions or opinions. When data is interpreted correctly, decisions become more accurate and reliable.

Example:
A school can improve academic performance by studying attendance records and exam results instead of making guesses.

2. Effective Use of Artificial Intelligence

Artificial Intelligence systems rely entirely on data. Data literacy ensures that data used to train AI systems is accurate, relevant, and unbiased, leading to better outcomes.

Example:
An AI system trained with quality data provides more accurate predictions and recommendations.

3. Improved Problem-Solving Skills

Data-literate individuals can recognize patterns, trends, and connections in data. This helps them approach problems logically and find effective solutions.

Example:
Traffic authorities analyze traffic flow data to reduce congestion by adjusting signal timings.

4. Awareness of Data Security and Privacy

Data literacy increases awareness about protecting personal information. It encourages responsible digital behavior and helps individuals understand the risks of sharing data online.

Example:
Students who understand data privacy avoid sharing personal details on unknown apps or websites.

5. Reduced Misinterpretation of Information

Without data literacy, information can easily be misunderstood or misused. Data literacy helps individuals interpret data correctly and avoid drawing false conclusions.

Example:
Understanding the source and size of a survey helps people avoid believing misleading statistics shown in advertisements or news reports.

Key Message:
Data literacy empowers individuals to question information, think critically, and make responsible decisions in a data-driven world.

2.1.3 How to Become Data Literate?

Every piece of data tells a story, but it is important to understand that not every story is true or complete. A data-literate person is someone who can interact with data thoughtfully in order to understand situations, make informed decisions, and avoid being misled.

Data literacy helps individuals observe, question, analyze, and interpret data instead of accepting information blindly. This skill is especially important in the digital world, where data is presented in many forms such as numbers, ratings, graphs, advertisements, and online reviews.

Understanding Data Literacy Through an Example

Scenario: Choosing an Online Learning Course

When selecting an online learning course from an educational platform, students come across a large amount of data such as course fees, duration, ratings, reviews, instructor details, and course content. Data literacy helps students examine this information carefully before enrolling in a course.

While choosing a course, students usually ask questions like:

- Which course is available at an affordable price?
- Which course is rated highest by learners?
- Does the course cover all the required topics and match my learning level?

A data-literate student studies this data logically by comparing multiple courses, reading genuine reviews, and checking course details instead of making decisions based only on advertisements or promotional offers.

How a Data-Literate Person Uses Data

A data-literate individual follows certain steps while interacting with data:

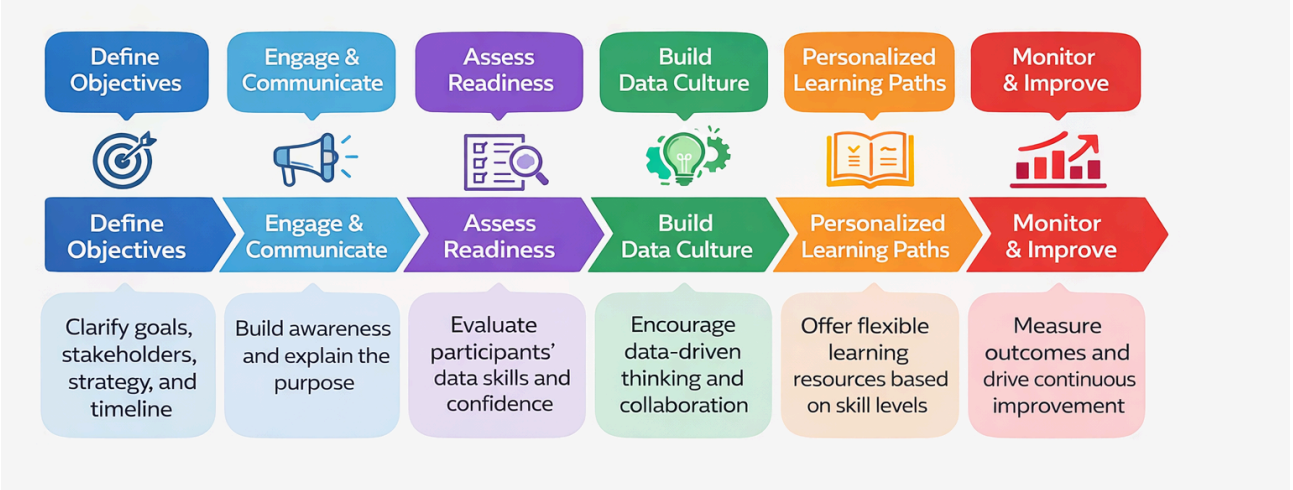
- Filters products according to requirements
  For example, if the budget is limited, the price filter can be set from *low to high* to find affordable options.
- Checks user ratings and reviews
  Products with higher ratings and genuine reviews are generally more reliable than those with no feedback.
- Analyzes product specifications
  The person verifies whether the video game meets specific needs such as platform compatibility, storage space, or multiplayer support.

- Compares multiple data sources
  Prices, reviews, and features are compared across different websites to avoid misleading information.

Data Literacy Process Framework

The Data Literacy Process Framework provides a structured approach to using data efficiently and responsibly. It helps individuals interact with data at every stage with awareness and critical thinking. This framework is iterative, meaning the steps can be repeated as new data becomes available.

The major stages of the framework are:



1. Define Objectives

The student first identifies the purpose of using data.

Example:
The objective is to select an online course that:

- Is affordable
- Matches the student's learning level
- Covers all required topics
- Has good learner ratings

2. Engage and Communicate

The student explores various online learning platforms and understands how courses are described.

Example:
Reading course descriptions, checking promotional offers, and understanding the learning outcomes mentioned by the platform or instructor.

3. Assess Readiness

At this stage, the student evaluates whether the available data is reliable and sufficient.

The student checks:

- Credibility of the platform
- Number and quality of reviews
- Recency of ratings

Example:
A course with many recent reviews is considered more trustworthy than one with very few ratings.

4. Build Data Culture

The student uses data-driven thinking by comparing different courses logically.

Example:
Comparing course fees, duration, ratings, and syllabus across multiple platforms to identify the best option.

5. Personalized Learning Paths

Based on analysis, the student selects a course that best suits their personal learning needs.

Example:
Choosing a course that may cost slightly more but offers better content coverage and higher learner satisfaction.

6. Monitor and Improve

After enrolling, the student reviews the outcome to improve future decisions.

Example:
If the course helps improve understanding and skills, the same data analysis approach can be used when selecting future courses.

Conclusion

By following the Data Literacy Process Framework, students can make informed and responsible decisions while choosing online courses. This approach helps avoid misleading information and ensures effective use of data in education and Artificial Intelligence.

Key Takeaway

> Using data step by step helps students choose the right course instead of relying on advertisements or assumptions.

2.1.4 What are Data Security and Privacy? How are they related to AI?

In the digital age, large amounts of data are created and shared every day. This data often includes **personal and sensitive information**. Therefore, it is very important to protect data and use it responsibly. Two key concepts related to this are **data security** and **data privacy**.

What is Data Security?

Data security refers to the methods and practices used to **protect data from unauthorized access, misuse, loss, or damage**. It ensures that data remains safe and is accessed only by authorized users.

Data security focuses on **how data is protected**.

Examples of data security measures include:

- Using strong passwords
- Encrypting data
- Installing antivirus software
- Restricting access to data
- Regularly updating systems

Example:
A school protects student records by allowing access only to teachers and administrators using secure login credentials.

What is Data Privacy?

Data privacy refers to the **right of individuals to control how their personal data is collected, used, and shared**. It focuses on respecting personal information and using it only for legitimate purposes.

Data privacy focuses on **who can use the data and why**.

Personal data may include:

- Name and address
- Phone number
- Email ID
- Photos and videos
- Location and online activity

Example:
An online app asking permission before accessing a user's location is an example of data privacy.

How are Data Security and Privacy Related to AI?

Artificial Intelligence systems **depend heavily on data** to learn, make predictions, and take decisions. Much of this data is personal or sensitive, making data security and privacy extremely important in AI systems.

1. AI Uses Large Amounts of Data

AI systems collect and process data such as:

- Search history
- Voice recordings
- Images and videos
- Health and education records

If this data is not protected, it can be misused.

Secure data leads to **accurate and reliable AI systems**.

Example:
Voice assistants use voice data to understand commands, so this data must be stored securely.

Secure data leads to **accurate and reliable AI systems**.

2. Data Security Protects AI Systems

Strong data security prevents:

- Data theft
- Unauthorized access
- Manipulation of AI models

Example:
If training data is altered, an AI system may give incorrect results.

3. Data Privacy Builds Trust in AI

When AI systems respect privacy, users feel safe sharing their data. This trust is essential for AI adoption.

Example:
A health AI system that protects patient records encourages people to use it without fear.

4. Ethical Use of AI

Data privacy ensures AI systems are used ethically and do not violate personal rights.

Example:
AI should not use personal photos or messages without permission.

Best Practices for Data Security and Privacy

To stay safe in an AI-driven digital world:

- Do not share personal information unnecessarily
- Use strong and unique passwords
- Read app permissions carefully
- Avoid clicking unknown links
- Log out from shared devices

Conclusion

Data security and data privacy are essential in today's AI-powered world. While data security protects data from misuse, data privacy ensures that personal information is used responsibly. Together, they help build **safe, trustworthy, and ethical AI systems**.

Key Takeaway

> AI can work effectively only when data is secure and personal privacy is respected.

Do's and Don'ts of Data Security and Data Privacy

✅ Do's

- **Do use strong passwords** that include letters, numbers, and symbols.
- **Do keep your passwords private** and change them regularly.
- **Do log out** from accounts after using shared or public computers.
- **Do check app permissions** before installing or using any application.
- **Do use secure websites** (look for "https") while sharing personal information.
- **Do update software and apps** regularly to fix security issues.
- **Do think before sharing** photos, personal details, or location online.
- **Do use antivirus software** to protect devices from malware.
- **Do verify sources** before trusting online information.
- **Do report suspicious messages or links** to a trusted adult or teacher.

❌ Don’ts (Unsafe Practices)

- **Don’t share personal information** like passwords, OTPs, or bank details.
- **Don’t click on unknown links** or pop-up advertisements.
- **Don’t use the same password** for multiple accounts.
- **Don’t accept friend requests** from unknown people on social media.
- **Don’t download apps** from untrusted sources.
- **Don’t share private photos or videos** on public platforms.
- **Don’t ignore privacy settings** on social media and apps.
- **Don’t believe all online offers** without checking authenticity.
- **Don’t leave devices unlocked** in public places.
- **Don’t store sensitive data** on shared or public devices.

Revision Time

1. Data literacy means the ability to:
   A. Store data
   B. Analyze and use data responsibly
   C. Create AI
   D. Collect data only
   Answer: B

2. Marks scored in an exam are an example of:
   A. Categorical data
   B. Text data
   C. Numerical data
   D. Image data
   Answer: C

3. Which is the lowest level of the Data Pyramid?
   A. Knowledge
   B. Wisdom
   C. Information
   D. Data
   Answer: D

4. Blood group belongs to which type of data?
   A. Numerical
   B. Text
   C. Image
   D. Categorical
   Answer: D

5. Data literacy helps in online shopping by:
   A. Increasing discounts
   B. Trusting all offers
   C. Identifying misleading information

D. Making ads
Answer: C

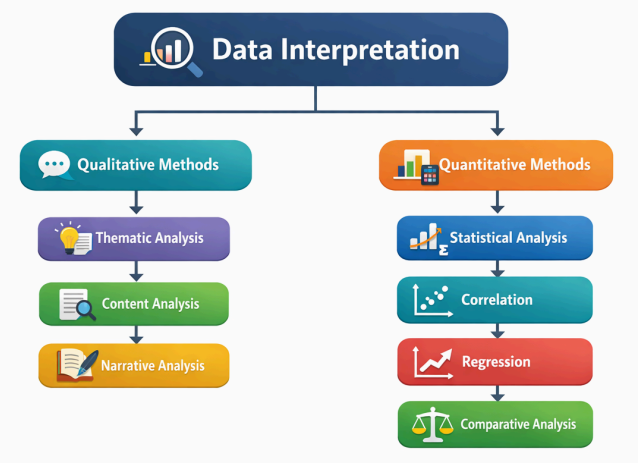
2.2 Acquiring Data, Processing, and Interpreting Data

In today's digital age, data is an important part of our daily life. Every time we use a mobile phone, check exam results, shop online, or browse social media, data is created and collected. This data helps individuals, organizations, and governments make decisions. However, raw data by itself is not very useful. It becomes meaningful only when it is collected correctly, processed carefully, analyzed for patterns, presented clearly, and interpreted properly. Incorrect or poorly handled data can lead to wrong conclusions and decisions.

Understanding the complete journey of data helps us develop data literacy, which allows us to think logically, understand information better, and make informed decisions in everyday life. The journey of data can be summarized in five key steps:

1. Data Acquisition – collecting data from different sources
2. Data Processing – organizing and cleaning the collected data
3. Data Analysis – examining processed data to find patterns and trends
4. Data Presentation – displaying data in understandable forms like tables, charts, and graphs
5. Data Interpretation – understanding what the data tells us and drawing useful conclusions

Once we understand these steps, it becomes easier to work with data effectively. The next important step is to understand the types of data, as knowing the type of data helps us decide how to analyze, process, and present it effectively.



Types of Data

Data can be broadly classified into two main types: Quantitative Data and Qualitative Data.

1. Quantitative Data

Quantitative data is numerical and can be measured, counted, or expressed in numbers. This type of data is useful for calculations, comparisons, and statistical analysis.

Features of Quantitative Data:

- Represented in numbers
- Can be added, subtracted, averaged, or graphed
- Useful for comparing quantities and identifying trends

Examples of Quantitative Data:

- Marks obtained by students in exams
- Height and weight measurements
- Attendance percentage of students
- Temperature readings of a city
- Number of books in a library

Types of Quantitative Data:

1. Discrete Data

Definition:
Discrete data consists of **countable, separate values**. These values usually come in whole numbers and cannot be divided into smaller parts meaningfully.

Features of Discrete Data:

- Values are **countable**
- Usually expressed in **whole numbers**
- No intermediate values exist between two consecutive points
- Often used for counting objects, people, or events

Examples of Discrete Data:

- Number of students in a class
- Number of cars in a parking lot.

2. Continuous Data

Definition:
Continuous data consists of **measurable values** that can take any value within a range. These values can be **fractions or decimals** and are not restricted to whole numbers.

Features of Continuous Data:

- Values are **measurable**
- Can take **any value within a given range**
- Often expressed in fractions or decimals
- Suitable for measurement rather than counting

Examples of Continuous Data:

- Height of students (e.g., 152.3 cm)
- Weight of fruits (e.g., 1.75 kg)

Importance in Analysis:
Quantitative data allows us to perform statistical operations such as mean, median, mode, and percentage. It is often presented using bar charts, line graphs, and histograms. In Artificial Intelligence, quantitative data is essential for prediction, pattern recognition, and decision-making.

2. Qualitative Data

Qualitative data describes qualities, characteristics, or categories rather than numbers. It is usually non-numerical and represents opinions, labels, or classifications.

Features of Qualitative Data:

- Non-numerical in nature
- Represents attributes or qualities
- Useful for classification and grouping

Examples of Qualitative Data:

- Student feedback on teaching methods
- Opinions in surveys like “satisfied” or “unsatisfied”
- Grades such as “Excellent,” “Good,” “Poor”
- Gender categories like male, female, or other
- Types of transport used: bus, train, bicycle

Importance in Analysis:

Qualitative data helps in grouping and categorizing information. It is often presented using tables, charts, or diagrams. In Artificial Intelligence, qualitative data is used for classification, sentiment analysis, recommendation systems, and customer feedback analysis.

Classification of Data Based on Collection

Data can also be classified based on **how it is collected**. Understanding this helps in determining its **accuracy, reliability, and suitability** for analysis. The two main types are **Primary Data** and **Secondary Data**.

1. Primary Data

Primary data is data that is **collected directly from the original source** for a specific purpose. It is fresh, original, and has not been previously recorded or used.
Primary data is collected firsthand to meet the needs of a particular study or research. Since it is gathered directly by the researcher, it is usually **more accurate and reliable** for the intended purpose. This type of data can be carefully designed to answer specific questions. Primary data is highly accurate and relevant because it is collected specifically for the study. It is also up-to-date and closely matches the research objective. Collecting primary data can be **time-consuming** and may involve extra effort and cost. Proper planning, resources, and careful execution are required to collect this data effectively.

Examples of Primary Data:
Primary data can be collected in many ways. For example, conducting a survey among students to find their favourite subject, interviewing teachers to understand classroom challenges, performing experiments to test a science hypothesis, or observing traffic flow at a busy road junction.

2. Secondary Data

Definition:
Secondary data is **collected by someone else** and reused for analysis. It is data that already exists and was collected for a different purpose.

Features of Secondary Data:

- Collected from existing sources such as books, articles, and online databases
- Easier and quicker to obtain
- May require verification for accuracy and relevance

Examples of Secondary Data:

- **Government reports** on population or health statistics
- **Research articles** published in journals
- **Online databases** and encyclopedias
- Previous surveys or studies conducted by other organizations

Advantages:

- Saves time and effort
- Often inexpensive or free to access

Limitations:

- May not fully match the specific needs of your study
- Accuracy and reliability must be checked

Classification of Data Based on Format

In the context of Artificial Intelligence (AI), data can also be classified based on its format and organization. Understanding the format of data is important because it determines how the data can be stored, processed, and analyzed. The two main types are Structured Data and Unstructured Data.

1. Structured Data

Definition:
Structured data is **organized and arranged in a fixed format**, making it easy to store, access, and analyze. This type of data fits neatly into **tables, rows, and columns**.

Features of Structured Data:

- Follows a **predefined format**

- Easy to search, process, and analyze
- Can be stored in **databases or spreadsheets**
- Compatible with standard data analysis tools

Examples of Structured Data:

- Attendance sheets of students
- Marks lists or grade books
- Employee records in a company
- Inventory lists in a store

Importance in AI:
Structured data can be directly fed into **AI algorithms** for analysis, predictions, and reporting. For example, a student performance prediction system can directly use marks and attendance records.

2. Unstructured Data

Definition:
Unstructured data is **not organized in a fixed format**. It does not fit easily into rows and columns, making it more difficult to process and analyze.

Features of Unstructured Data:

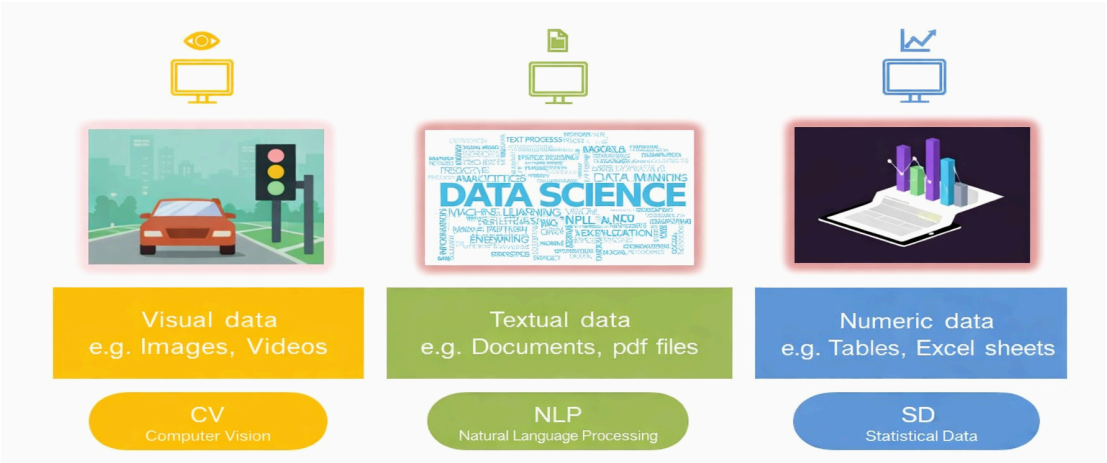
- No predefined format
- Can be text, audio, video, or images
- Requires specialized tools and methods for processing
- Often used for advanced AI tasks like pattern recognition and sentiment analysis

Examples of Unstructured Data:

- Images and photographs
- Video recordings from CCTV or lectures
- Audio files such as podcasts or voice notes
- Social media posts, comments, and reviews

Importance in AI:
AI systems need to **process unstructured data carefully** using techniques like **Natural Language Processing (NLP)** for text or **Computer Vision** for images and videos. Once processed, unstructured data can be analyzed to extract meaningful insights.

Types of Data Utilized Across the Three Major Domains of Artificial Intelligence



Features of Data

Data has several important features that determine its quality and usefulness. First, data should be accurate, meaning it should be correct and free from errors, because inaccurate data can lead to wrong conclusions. Data should also be complete, as missing information can reduce its value and affect results. Consistency is another important feature of data; the same data should remain the same when used in different systems or records. Data must be timely, which means it should be available when needed and should be up to date. Relevance is essential because only data related to a particular purpose or problem is useful, while unnecessary data creates confusion. Reliable data is collected from trusted and authentic sources and can be depended upon for decision-making. Validity ensures that data follows proper rules, formats, and limits, such as numbers falling within an acceptable range. Data should also be unique so that each record represents a single item or person without duplication. Lastly, data should be accessible and easy to retrieve so that users can use it whenever required. All

these features together help in maintaining good-quality data and make it meaningful for study, analysis, and decision-making

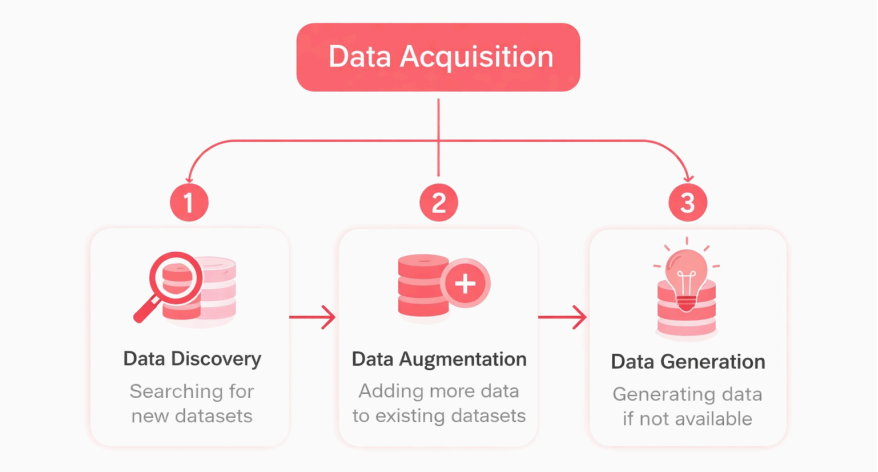
What is Data Acquisition?

Data acquisition is the process of collecting data for a specific purpose. The collected data may come from different sources depending on the situation. In schools, data can be collected from attendance registers, examination records, surveys, or online forms. In Artificial Intelligence systems, data is collected so that machines can learn patterns and make predictions.

The quality of an AI system depends heavily on the quality of the data it receives. Accurate, relevant, and sufficient data helps produce reliable results, while poor-quality data may lead to incorrect conclusions.

Examples of data acquisition include:

- Collecting students' marks for performance analysis
- Recording daily temperature for weather study
- Gathering feedback through online surveys



Improving Library Book Management :

- **Goal:** Understand which books are popular and which are rarely used.
- **Requirement:** Data on book issues, student feedback, and catalog searches.
- **Process:** The library collected this data and used **data discovery techniques** to analyze patterns:
  Which books were borrowed most frequently
  Which books were never borrowed
  Peak times for book borrowing

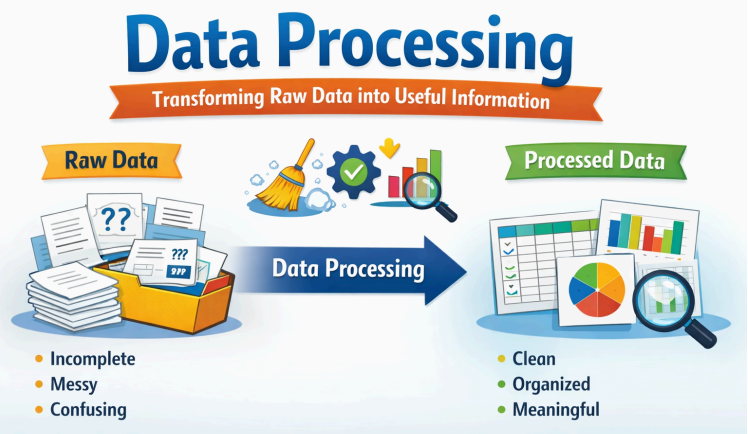
- **Outcome / Insights:**
  Order more copies of popular books
  Remove rarely used books
  Improve library services based on student needs

This process of exploring and analyzing collected data to find useful patterns is called **data discovery**.

What is Data Processing?

Data processing is the step in which raw data is transformed into a clear, organized, and useful form. Raw data is usually collected from different sources and is often incomplete, repetitive, or confusing. In this form, data cannot be easily understood or used for decision-making. Data processing helps improve the quality of data so that it becomes ready for analysis.

Data processing means cleaning and organizing data so that it makes sense.



Why is Data Processing Necessary?

Data processing is important because:

- Raw data may contain mistakes or missing values
- Data may be collected in different formats
- Large amounts of data are difficult to understand directly
- Incorrect data can lead to wrong conclusions

By processing data, we ensure accuracy, consistency, and clarity, which are essential for reliable results, especially in Artificial Intelligence systems.

Steps Involved in Data Processing

Data processing usually involves the following steps:

1. Data Cleaning

Data cleaning means removing errors, duplicates, and incorrect entries from the data. This step improves the accuracy of data.

Example:
If a student's marks are entered twice or incorrectly, the extra or wrong entry is removed.

2. Data Organization

In this step, data is arranged in a proper structure such as rows and columns. Organized data is easier to read and compare.

Example:
Arranging students' marks according to subjects or roll numbers in a table.

3. Data Classification

Data classification involves grouping similar data together based on certain criteria.

Example:
Grouping students based on grade ranges such as below average, average, and above average.

4. Data Transformation

Sometimes data needs to be converted into a suitable format so that it can be used for analysis.

Example:
Converting attendance records from written text into numerical percentages.

5. Data Storage

After processing, data is stored in computers, databases, or spreadsheets for future use.

Example:
Saving processed exam results in a school database.

Example of Data Processing in Daily Life

Suppose a teacher collects raw marks of students written randomly on paper. These marks do not clearly show performance. During data processing, the teacher:

- Removes incorrect entries
- Arranges marks student-wise
- Calculates totals and averages
- Stores the final data in a result sheet

Role of Data Processing in Artificial Intelligence

In Artificial Intelligence, data processing is a very important step. AI systems cannot work with raw data directly. The data must first be cleaned and organized so that machines can learn patterns correctly.

Example:
If an AI system is trained to predict student performance, incorrect or unprocessed data may lead to wrong predictions.

Key Point to Remember

Data processing turns raw, confusing facts into structured and meaningful information. Without proper data processing, analysis and interpretation may give misleading results.

What is Data Analysis?

Data analysis is the process of carefully examining processed and organized data to discover useful information. It helps us understand **patterns, trends, and relationships** hidden within data. While raw or processed data only shows numbers or facts, data analysis explains **what the data means** and **why certain results occur**. This step is very important because it turns data into knowledge that can support decisions.

In simple terms, **data analysis helps us make sense of data**.

Why is Data Analysis Important?

Data analysis is important for several reasons:

- It helps in understanding large amounts of data easily
- It allows comparison between different sets of data
- It helps in identifying changes over time
- It supports accurate and fair decision-making

Without proper analysis, data may remain unused or may lead to incorrect conclusions.

Common Activities in Data Analysis

Data analysis involves different activities such as:

- **Calculating values** like totals, averages, or percentages
- **Comparing data** to identify differences or similarities
- **Finding patterns** such as regular improvement or decline
- **Identifying trends** such as increase or decrease over time

These activities help answer important questions like:

- What is happening?
- Why is it happening?
- What can be learned from this data?

Examples of Data Analysis in School Life

1. **Average Marks of a Class**
   By calculating the average marks, teachers can understand the overall performance of the class.

2. **Subject-wise Performance**
   By comparing marks in different subjects, it becomes clear which subjects students find easier or more difficult.

3. **Attendance Analysis**
   By studying attendance data over several months, schools can identify patterns such as frequent absences during certain periods.

Role of Data Analysis in Artificial Intelligence

In Artificial Intelligence, data analysis plays a key role in helping machines learn from data. AI systems analyze past data to recognize patterns and use them to make predictions or decisions.

Examples:

- An AI system analyzes students' past marks to predict future performance
- Online platforms analyze user behavior to recommend videos or courses
- Weather systems analyze historical data to forecast weather conditions

Accurate analysis helps AI systems improve over time and produce reliable results.

Basis	**Data Processing**	**Data Analysis**
Meaning	Data processing is the step of cleaning, organizing, and preparing raw data.	Data analysis is the step of examining processed data to find patterns, trends, and relationships.
Purpose	To convert raw data into a usable and structured form.	To understand what the data shows and why certain results occur.
Nature of Work	Focuses on accuracy, arrangement, and consistency of data.	Focuses on interpretation, comparison, and drawing conclusions.
Type of Output	Produces organized data such as tables or lists.	Produces insights, patterns, and meaningful information.
Example	Arranging students' marks subject-wise in a table.	Finding the average marks or identifying the best-performing subject.
Role in AI	Prepares clean and correct data for AI systems.	Helps AI systems learn from data and improve predictions.

Data processing turns raw, confusing facts into structured and meaningful information. Without proper data processing, analysis and interpretation may give misleading results.

What is Data Presentation?

Data presentation is the process of showing data in a clear, organized, and easy-to-understand manner. When data is properly presented, it becomes easier to read, compare, and analyze. Raw numbers written randomly may confuse readers, but when the same data is presented visually or systematically, it communicates information effectively.

The main purpose of data presentation is to **make data meaningful and understandable for others**.

Why is Data Presentation Important?

Data presentation is important because:

- It simplifies large amounts of data.
- It helps compare values easily.
- It saves time while understanding information.
- It supports better analysis and interpretation.
- It makes communication of data more effective.

Well-presented data helps students, teachers, and decision-makers quickly understand results and trends.

Common Methods of Data Presentation

Data can be presented in different forms depending on the type of data and purpose.

1. Tables

Tables are a simple and systematic way of presenting data. In a table, information is arranged in **rows and columns**, which makes it easy to read, compare, and understand. Tables are especially useful when **exact values and details** are important.

Each row usually represents an item or individual, while each column represents a particular category or variable. Because of this clear structure, tables help avoid confusion and allow quick reference to specific data.

Example:

A table showing the marks obtained by students in different subjects helps teachers and students compare performance across subjects and identify strengths or weaknesses easily.

Marks of Students in Subjects

Student Name	Mathematics	Science	English
Rahul	78	82	75
Anjali	85	88	90
Aman	70	74	72

2. Bar Graphs

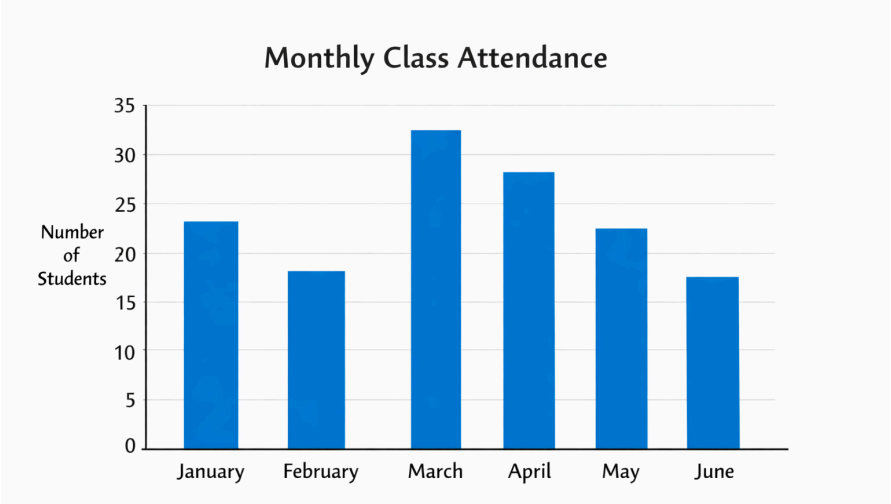
Bar graphs are used to **compare values among different categories**. They represent data using **rectangular bars** of equal width, where the height or length of each bar shows the value of the data.

Bar graphs make it easy to **observe differences, similarities, and patterns** at a glance. They are especially useful when comparing discrete data such as groups, classes, or time periods.

The bars can be drawn either **vertically or horizontally**, depending on the type of data being presented. A scale is used to ensure accurate representation of values.

Example:

A bar graph showing the **monthly attendance of a class** helps in comparing attendance levels across different months and identifying months with higher or lower attendance.



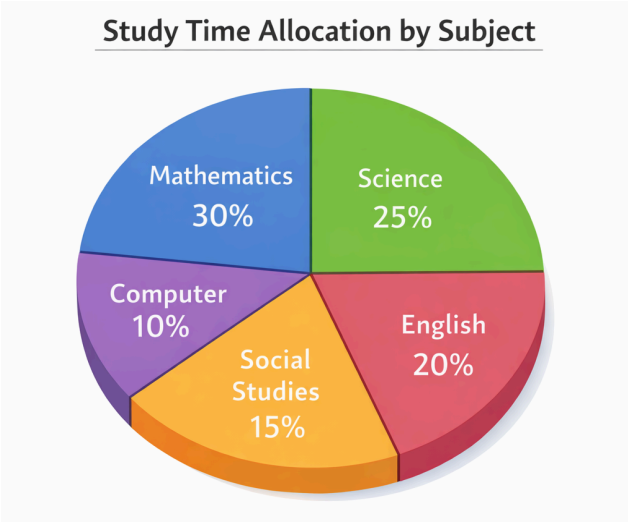
3. Pie Charts

Pie charts present data using a **circular diagram** that is divided into **sectors or slices**. Each slice represents a part of the whole, making pie charts especially useful for showing **percentages, ratios, or proportions**.

The entire circle represents **100% of the data**, and the size of each slice depends on its value. Pie charts help students quickly understand how different categories contribute to the total.

Example:

A pie chart showing how students **divide their study time among different subjects** helps identify which subjects receive more or less attention



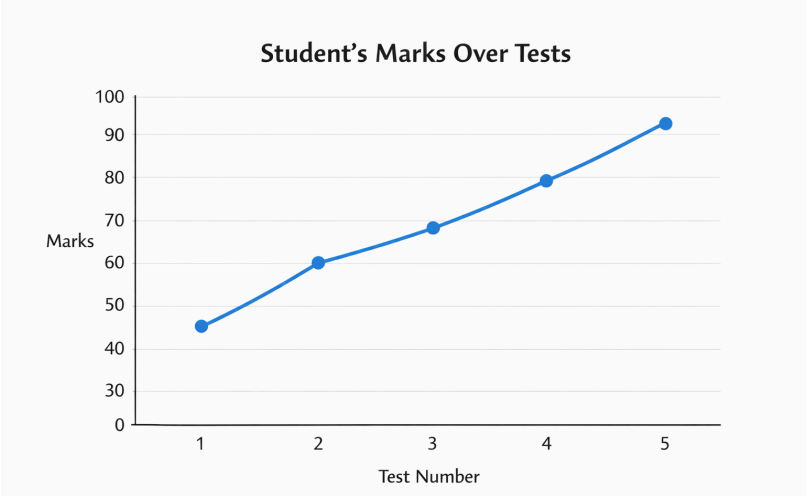
4. Line Graphs

Line graphs are used to **show changes in data over a period of time**. They display data points connected by straight lines, making it easy to observe **trends, increases, decreases, or patterns**.

The horizontal axis usually represents **time** (such as days, months, or tests), while the vertical axis represents the **measured values**. Line graphs are especially helpful when tracking progress or performance.

Example:

A line graph showing the **improvement in a student's marks across several tests** helps understand learning progress and performance trends over time.



Role of Data Presentation in Artificial Intelligence

In Artificial Intelligence, **data presentation plays an important role in understanding the results produced by AI systems**. AI models process large amounts of data and identify patterns that may not be easily visible in raw form. Presenting this data in a visual way helps both developers and users understand how the system is working.

Graphs, charts, and dashboards are commonly used to display AI outputs. These visual tools make complex results **easy to read, compare, and explain**. For example, a graph can show how an AI model's accuracy improves over time, while a dashboard can display predictions, trends, or alerts at a glance.

Effective data presentation improves **decision-making, transparency, and trust in AI systems**, as users can clearly see and interpret the patterns discovered by the AI.

What is Data Interpretation?

Data interpretation is the process of understanding and explaining the meaning of data after it has been collected, analyzed, and presented. It focuses on drawing logical conclusions, identifying what the results indicate, and using those findings to make informed decisions.

Data interpretation answers the question: *"What does this data tell us?"* It helps convert numbers, charts, or tables into useful information that can guide actions.

For example, if a chart shows that students who study regularly score higher marks, data interpretation helps us conclude that consistent study habits improve academic performance. Without interpretation, data remains just numbers without meaning.

In Artificial Intelligence, data interpretation is essential because it helps developers and users understand AI predictions and outcomes, check whether results are reasonable, and decide what steps should be taken next.

Data interpretation is important because it transforms raw facts and analyzed data into **useful knowledge**. Numbers, tables, and graphs by themselves do not clearly explain what is happening. Interpretation helps us understand the meaning behind the data and what conclusions can be drawn from it.

It plays a key role in **decision-making**. When data is correctly interpreted, it allows individuals, schools, organizations, and governments to take well-informed and logical decisions instead of relying on guesses or assumptions. For example, interpreting exam result data helps teachers decide where students need extra support.

Data interpretation also helps **prevent misunderstandings and incorrect conclusions**. The same data can be misleading if it is not carefully explained. Proper interpretation ensures that results are understood in the right context and reduces the chances of misusing information.

In addition, data interpretation helps **explain the reasons behind results**. It allows us to understand why a particular trend, pattern, or change has occurred, rather than just observing that it has happened.

Without data interpretation, charts, graphs, and figures remain only **numbers and shapes without meaning**. Interpretation gives purpose to data and helps turn information into meaningful insights that can be applied in real life.

Examples of Data Interpretation

- Data interpretation helps us understand what the data is trying to tell us and guides us toward meaningful conclusions.
- If the data shows that students score higher marks after regular practice, it can be interpreted that **consistent study habits lead to better academic performance**. This understanding encourages students to practice regularly and helps teachers plan effective study schedules.
- If attendance data shows a decrease during certain months, it may be interpreted that **seasonal factors, weather conditions, or health-related issues** are affecting student attendance. Schools can then take suitable steps, such as awareness programs or schedule adjustments.

- If data indicates that students perform better in one subject compared to others, it may suggest that **the teaching methods or learning resources used for that subject are effective**. This insight can help teachers apply similar methods to other subjects.
- These examples show how data interpretation turns observations into **useful actions and informed decisions**.

Role of Data Interpretation in Artificial Intelligence

Data interpretation plays a very important role in Artificial Intelligence because it allows AI systems to **understand analyzed data and act on it meaningfully**. After data is collected, processed, and analyzed, AI systems interpret the results to produce useful outputs. Artificial Intelligence systems use data interpretation to **make predictions**. By studying past data patterns, AI can predict future outcomes, such as forecasting weather conditions or estimating a student's academic performance.Data interpretation also helps AI systems **give recommendations**. By understanding user behavior and preferences, AI can suggest suitable content or products. For example, an AI system recommends videos, songs, or products based on a user's previous interests and viewing history. Another key role of data interpretation in AI is to **support decision-making**. AI systems analyze large amounts of data and interpret the results to help humans make better and faster decisions.

Examples:

An AI-based video platform recommends videos based on the user's interests and viewing patterns.

A health AI system interprets medical data to predict the risk of diseases and assist doctors in diagnosis.

A learning app analyzes student performance data and suggests topics or practice areas where improvement is needed.

Thus, data interpretation enables Artificial Intelligence systems to turn data into **intelligent actions and useful insights**.

Relationship Between Data Presentation and Data Interpretation

Data presentation and data interpretation are closely connected and depend on each other. Data presentation focuses on displaying data clearly, using tables, charts, graphs, or diagrams so that information is easy to understand. Data interpretation focuses on explaining what the presented data means and what conclusions can be drawn from it. Proper data presentation

makes interpretation easier and more accurate. When data is arranged neatly and shown in a suitable visual form, patterns, trends, and comparisons become clear. This helps readers understand the information quickly and correctly. If data is poorly presented or confusing, interpretation can become incorrect or misleading. For example, an unclear graph or wrongly labelled chart may lead to false understanding of the data. Therefore, both data presentation and data interpretation must be carried out carefully and responsibly so that information is understood in the right way and used for correct decision-making.

Activity-Based Understanding

Activity-based learning helps students clearly understand the connection between data presentation and data interpretation. Students can begin by collecting simple data, such as the number of hours they study each day. This data can then be presented using a table, bar graph, or line graph so that the information becomes easy to read and compare.

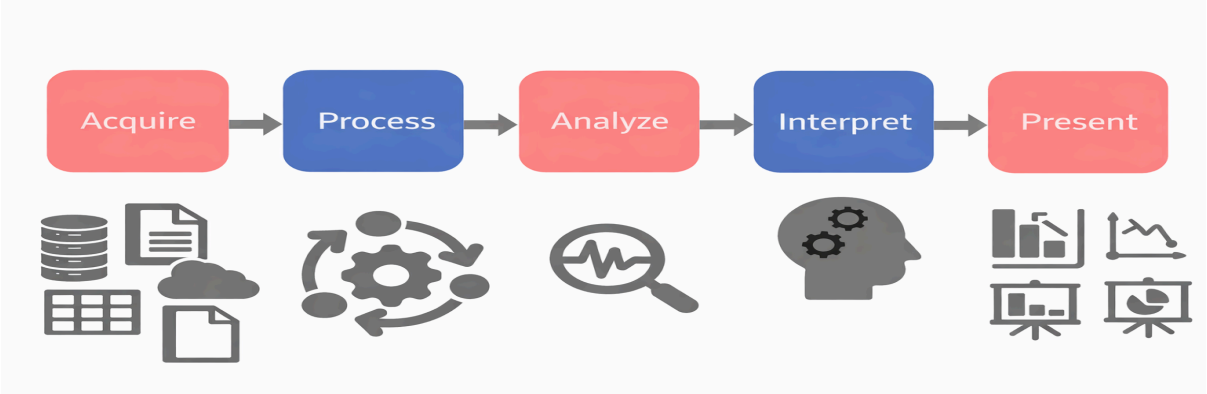
After presenting the data, students interpret the results by observing patterns and trends. For example, they may notice that study hours increase before exams or that regular study leads to better performance. Through this process, students learn how raw numbers gradually turn into meaningful conclusions.

Data presentation helps organize and display data in a clear and simple form, while data interpretation explains what the data actually means. When used together, they help individuals—and even Artificial Intelligence systems—understand information correctly and make informed and responsible decisions.

Step	Meaning	Purpose	Key Activities	Example
Data Acquisition	Collecting raw data from various sources	To gather relevant information for a specific purpose	Surveys, observations, online forms, sensors	Collecting students' marks or attendance records
Data Processing	Organizing and cleaning collected data	To prepare data for analysis by	Cleaning errors, removing duplicates,	Arranging students' marks subject-wise and calculating totals

		removing errors and arranging it	organizing into tables, formatting	
Data Analysis	Examining processed data to find patterns, trends, or relationships	To understand what the data shows and why	Calculating averages, identifying trends, comparing groups	Finding average marks of a class or subjects where students perform better
Data Presentation	Displaying data in a clear and understandable way	To make data easy to read, compare, and interpret	Using tables, graphs, pie charts, and dashboards	Showing class marks in a bar graph or pie chart
Data Interpretation	Explaining the meaning of analyzed data	To draw conclusions and make informed decisions	Explaining trends, drawing conclusions, recommending actions	Concluding that regular study improves student performance

Data becomes meaningful only when it is collected accurately, processed in an organized manner, analyzed thoughtfully, and interpreted logically. Each step plays an important role in turning raw facts into useful information. When students understand this complete process, they learn to use data responsibly, think critically, and avoid being misled by incorrect or incomplete information. This understanding also prepares them to interact effectively with Artificial Intelligence systems, which depend on well-handled data to function correctly and provide reliable results.



Activity Guidelines

The table below shows the details of a school survey conducted on **daily screen time usage** of **45 students** for different purposes: Online Classes, Social Media, Gaming, Educational Videos, and Entertainment.

Student Screen Time Usage (in Hours per Day)

Screen Time Range (hrs)	Online Classes	Social Media	Gaming	Educational Videos	Entertainment
Less than 1	2	5	6	4	3
Between 1–2	10	8	9	12	7
Between 2–3	18	14	13	15	16
Between 3–4	9	11	10	8	12
4 and Above	6	7	7	6	7
Total Students	**45**	**45**	**45**	**45**	**45**

Tasks

Copy the above table into an Excel sheet and perform the following activities:

1. Create a **bar graph** showing the screen time distribution for all five purposes
2. Create a **pie chart** showing the screen time distribution for **Social Media**
3. Create a **line chart** showing the screen time distribution for **Gaming**

Revision Time :

1. Which step converts raw data into a usable form?

A. Data acquisition

B. Data processing

C. Data presentation

D. Data interpretation

Answer: B

2. Data that is numerical in nature is called:

A. Qualitative data

B. Secondary data

C. Quantitative data

D. Unstructured data

Answer: C

3. Which data type includes values like height and weight?

A. Discrete data

B. Continuous data

C. Primary data

D. Structured data

Answer: B

4. Which chart is best for showing parts of a whole?

A. Bar graph

B. Line graph

C. Table

D. Pie chart

Answer: D

5. Which step explains the meaning of analyzed data?

A. Data processing

B. Data analysis

C. Data acquisition

D. Data interpretation

Answer: D